

DIP-SMC-PSO: Technical Progress Memo

Contents

Model choices, algorithm design, and
experimental results

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Program: M.Sc. Program

February 2026 — Status: Research complete,
thesis 90%

Executive Summary

Problem. Stabilise a double-inverted pendulum (DIP) under model uncertainty and external disturbances using a single actuator ($u \in [-150, 150]$ N).

Approach. Four SMC variants (Classical, Super-Twisting, Adaptive, Hybrid Adaptive) PSO-tuned across 1,300+ simulations: 400 Monte Carlo runs, 15-scenario robust PSO, boundary layer study, and parametric uncertainty analysis.

Top result. STA-SMC: **1.82 s** settling time (−16% vs. Classical), **2.3%** overshoot, and **74%** chattering reduction (FFT metric). All three quantified improvements are statistically significant ($p \leq 0.038$; Cohen $d \geq 2.1$). MT-8 disturbance logs are retained as qualitative stress-test evidence only, because all controller-scenario entries are marked non-converged.

Critical open issues. (1) Robust PSO returned STA gains ($K_1=2.02$, $K_2=6.67$) violating the Moreno-Osorio condition $K_1 > K_2$ (Finding #20; unresolved—see Appendix ??). (2) Hybrid Adaptive controller returned sentinel values on all 100 Monte Carlo runs (mode-switching instability; retained for theoretical completeness only).

Thesis status. 90/100 score. January 2026 completion phases slipped due to ongoing research obligations. Revised target: **April 2026** (see updated timeline in Section ??).

Contributions

- **C1.** First systematic PSO-tuned boundary layer adaptation study on a 2-link DIP; includes honest correction of the 66.5% $\rightarrow$ 3.7% chattering reduction claim.
 - **C2.** Robust multi-scenario PSO with worst-case fitness reduces the generalisation gap from $50.4\times$ to $7.5\times$ on challenging (± 0.30 rad) initial-condition distributions.
 - **C3.** Simultaneous Lyapunov proofs (9 theorems, 5 lemmas) for all four SMC variants on the same DIP system, enabling direct stability-type comparison.
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1 System Model: Equations and Parameter Choices

1.1 State Vector and Coordinate Definition

- **DIP state vector:**

$$\mathbf{x} = [x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]^\top \in \mathbb{R}^6 \quad (1)$$

- **Coordinate convention:**

$$\mathbf{q} = [x, \theta_1, \theta_2]^\top, \quad \dot{\mathbf{q}} = [\dot{x}, \dot{\theta}_1, \dot{\theta}_2]^\top \quad (2)$$

Units: x [m], θ_i [rad], $\dot{x}$ [m/s], $\dot{\theta}_i$ [rad/s]. Sign convention: $+x$ rightward, $+\theta_i$ counterclockwise from upright.

- **Target equilibrium and error:**

$$\mathbf{x}^* = [0, 0, 0, 0, 0, 0]^\top, \quad \mathbf{e} = \mathbf{x} - \mathbf{x}^* \quad (3)$$

Tracking case: shift only the first component of $\mathbf{x}^*$.

- **Measured output:**

$$\mathbf{y} = [x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]^\top \quad (4)$$

Feedback source: full-state in simulation; encoder/estimator pipeline in HIL. Safety limits: benchmark configuration files.

1.2 Equations of Motion

- **Full nonlinear dynamics:**

$$\mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \mathbf{F}_{\text{fric}} = \mathbf{B} \mathbf{u} \quad (5)$$

with $\mathbf{q} = [x, \theta_1, \theta_2]^\top$, $\mathbf{B} = [1, 0, 0]^\top$, and actuator bounds $u \in [-150, 150]$ N.

- **Modeling assumptions:** planar rigid-body dynamics, ideal revolute joints, single-input bounded actuation, viscous + Coulomb friction, and disturbance injection through benchmark profiles (MT-8/LT-6).
- **Equivalent first-order form for control design:**

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x}) u + \mathbf{d}(t) \quad (6)$$

Disturbance term: $\mathbf{d}(t) = 0$ (nominal), $\mathbf{d}(t) \neq 0$ (rejection tests).

- **Structural properties used in proofs:** Properties: $\mathbf{M}(\mathbf{q})$ symmetric positive definite; $\dot{\mathbf{M}}(\mathbf{q}) - 2\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ skew-symmetric on admissible configurations.
- **Model partition in this workflow:**
 - Full model: final benchmark reporting.
 - Simplified model: PSO loops (reduced coupling, no gyroscopic term).

Inertia matrix $\mathbf{M}(\mathbf{q})$ (3×3 , symmetric, configuration-dependent):

- Compact notation: $c_1 = \cos \theta_1$, $c_2 = \cos \theta_2$, $c_{12} = \cos(\theta_1 - \theta_2)$.

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} M + m_1 + m_2 & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{bmatrix} \quad (7)$$

- Matrix entries:

$$\begin{aligned} a_{12} &= (m_1 l_{c1} + m_2 l_1) c_1 + m_2 l_{c2} c_2 \\ a_{13} &= m_2 l_{c2} c_2 \\ a_{22} &= m_1 l_{c1}^2 + m_2 (l_1^2 + l_{c2}^2) + I_1 + I_2 + 2m_2 l_1 l_{c2} c_{12} \\ a_{23} &= m_2 l_{c2}^2 + I_2 + m_2 l_1 l_{c2} c_{12} \\ a_{33} &= m_2 l_{c2}^2 + I_2 \end{aligned} \quad (8)$$

- Dominant coupling: a_{22} term $2m_2 l_1 l_{c2} c_{12}$; grows with large relative-link deflection.

Coriolis/centrifugal matrix $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ (non-zero entries):

- Computed from Christoffel symbols of (??); zero entries omitted.

$$\begin{aligned} C_{01} &= -(m_1 l_{c1} + m_2 l_1) \sin \theta_1 \dot{\theta}_1 - m_2 l_{c2} \sin \theta_2 \dot{\theta}_2 \\ C_{02} &= -m_2 l_{c2} \sin \theta_2 \dot{\theta}_2 \\ C_{11} &= C_{12} = -m_2 l_1 l_{c2} \sin(\theta_1 - \theta_2) \dot{\theta}_2 \\ C_{21} &= +m_2 l_1 l_{c2} \sin(\theta_1 - \theta_2) \dot{\theta}_1 \end{aligned} \quad (9)$$

- Indices $(0, 1, 2)$ correspond to (x, θ_1, θ_2) .
- Gyroscopic augmentation: $g_c(\dot{\theta}_1 + \dot{\theta}_2)$ added to C_{01} and $-C_{10}$; coefficient 0.01 (config-enabled).

Gravity and friction model:

$$\begin{aligned} \mathbf{G}(\mathbf{q}) &= \begin{bmatrix} 0 \\ G_2 \\ G_3 \end{bmatrix} \\ G_2 &= -(m_1 l_{c1} + m_2 l_1) g \sin \theta_1 \\ G_3 &= -m_2 l_{c2} g \sin \theta_2 \end{aligned} \quad (10)$$

- $G_1 = 0$ because gravity is perpendicular to cart translation.

$$\mathbf{F}_{\text{fric},i} = -b_i \dot{q}_i - \mu_i \text{sign}(\dot{q}_i) \mathbb{1}_{|\dot{q}_i| > 10^{-6}} \quad (11)$$

- Friction scope: all DOFs $i \in \{x, \theta_1, \theta_2\}$; dead zone $|\dot{q}_i| > 10^{-6}$ avoids sign discontinuity at rest.
- Viscous coefficients (0.2, 0.005, 0.004); Coulomb coefficients from config, Monte Carlo variation $\pm 10\%$.
- Simulation model: full viscous + Coulomb form (??); PSO speedup uses reduced coupling (0.7–0.8) and no gyroscopic term.
- Linearization: numerical $(\mathbf{A}, \mathbf{B})$ at $\mathbf{q}^* = \mathbf{0}, \dot{\mathbf{q}}^* = \mathbf{0}$ with $\varepsilon = 10^{-8}$; $\Delta t = 0.01$ s, $u \in [-150, 150]$ N.

1.3 Physical Parameters

Parameter	Symbol	Value
Cart mass	M	1.5 kg
Link 1 mass	m_1	0.2 kg
Link 2 mass	m_2	0.15 kg
Link 1 length	l_1	0.4 m
Link 2 length	l_2	0.3 m
Link 1 COM distance	l_{c1}	0.2 m
Link 2 COM distance	l_{c2}	0.15 m
Link 1 inertia	I_1	0.0081 kg·m ²
Link 2 inertia	I_2	0.0034 kg·m ²
Gravity	g	9.81 m/s ²
Cart friction	b_c	0.2 N·s/m
Joint 1 friction	b_1	0.005 N·m·s/rad
Joint 2 friction	b_2	0.004 N·m·s/rad
Max control force	$u_{\max}$	150 N
Control period	Δt	0.01 s (100 Hz)

1.4 Model Selection: Simplified vs. Full Non-linear

Model	Use in this project	Rationale
Simplified (linearized around upright)	PSO optimization (all controllers)	10–50× faster per simulation step. PSO requires thousands of evaluations; full model would be prohibitive.
Full nonlinear (??)	All published benchmark results	Complete Lagrangian, all coupling terms, Coriolis, centrifugal, gyroscopic, full friction. Used for MT-5, MT-7, MT-8, LT-6.
Low-rank (SVD approximation)	HIL mode (hardware-in-the-loop)	Reduced singular-value truncation preserves stability with lower real-time compute load.

- **Decision basis:** full nonlinear PSO ($40 \times 200 \times 10$ s) exceeds 8 h/controller.
- **Optimization runtime:** minutes on simplified model.
- **Reporting model:** full nonlinear only.

Simplification summary (Section 1.4, PSO only).

- **Baseline kept:** same state and actuation structure as (??) and (??), with unchanged bounds $u \in [-150, 150]$ N and $\Delta t = 0.01$ s.
- **Linearization step:** numerical $(\mathbf{A}, \mathbf{B})$ around upright equilibrium $(\mathbf{q}^*, \dot{\mathbf{q}}^*) = (\mathbf{0}, \mathbf{0})$ using $\varepsilon = 10^{-8}$.
- **Term reduction:** coupling terms scaled down (0.7–0.8 range) and gyroscopic augmentation disabled during PSO loops.
- **Scope guard:** all final figures/tables use the full nonlinear model; simplification is used only to reduce PSO search runtime (about 10–50×).

2 Sliding Mode Control: Surface Design and Control Laws

2.1 Sliding Surface

- **Compact state form:**

$$\sigma = \lambda_1(\theta_1 - \theta_1^*) + \lambda_2(\theta_2 - \theta_2^*) + k_1(\dot{\theta}_1 - \dot{\theta}_1^*) + k_2(\dot{\theta}_2 - \dot{\theta}_2^*) \quad (12)$$

- **Equivalent error form** ($e_i = \theta_i - \theta_i^*$, $\theta_i^* = 0$):

$$\sigma = \lambda_1 e_1 + \lambda_2 e_2 + k_1 \dot{e}_1 + k_2 \dot{e}_2 \quad (13)$$

- **Sliding dynamics, stability, and tuning:**

$$\sigma = 0 \Rightarrow \dot{e}_i = -\frac{\lambda_i}{k_i} e_i, \quad \lambda_i > 0, k_i > 0, i \in \{1, 2\} \quad (14)$$

Time constants: $\tau_i = k_i/\lambda_i$ (direct tuning meaning). MT-8 robust Classical example: $(\lambda_1, \lambda_2, k_1, k_2) = (5.51, 3.49, 23.07, 12.85)$ gives $\tau_1 \approx 4.19$ s and $\tau_2 \approx 3.68$ s.

- **Units and scaling:** Signal units: θ_i [rad], $\dot{\theta}_i$ [rad/s]. Scaling policy: no state normalization before PSO; scaling handled by gain bounds and cost normalization.

- **Relative degree, robustness, and practical limitation:**

$$\dot{\sigma} = \lambda_1 \dot{\theta}_1 + \lambda_2 \dot{\theta}_2 + \mathbf{L}\mathbf{M}^{-1}(\mathbf{B}\mathbf{u} - \mathbf{F}) + \mathbf{L}\mathbf{M}^{-1}\mathbf{B}d(t), \quad \mathbf{L} = [0, k_1, k_2], \quad \mathbf{F} = \mathbf{C}\dot{\mathbf{q}} + \mathbf{G} + \mathbf{F}_{\text{fric}}. \quad (15)$$

Reachability: select $K > \bar{d} + \eta$ when $|d(t)| \leq \bar{d}$. Limitation: surface excludes cart x ; mitigated by saturation $u \in [-150, 150]$ N and optional outer-loop cart reference shaping.

2.2 Controller Variants: Control Laws

- **Unified template (all variants):**

$$u = u_{\text{eq}} + u_{\text{sw}} + u_{\text{aux}} \quad (16)$$

Mapping: variant-specific choices of $(u_{\text{eq}}, u_{\text{sw}}, u_{\text{aux}})$.

2.2.1 Classical SMC: Saturation Function and Equivalent Control

- **Saturation function:**

$$\text{sat}\left(\frac{\sigma}{\varepsilon}\right) = \begin{cases} \sigma/\varepsilon & |\sigma| \leq \varepsilon \\ \text{sign}(\sigma) & |\sigma| > \varepsilon \end{cases} \quad (17)$$

- **Control law mapping to (??):**

$$u = \underbrace{u_{\text{eq}}}_{\text{equivalent}} + \underbrace{K \text{sat}\left(\frac{\sigma}{\varepsilon}\right)}_{\text{switching}} + \underbrace{-k_d \dot{\sigma}}_{\text{auxiliary damping}} \quad (18)$$

- **Equivalent control term (from $\dot{\sigma} = 0$):**

$$u_{\text{eq}} = -\frac{\mathbf{L}\mathbf{M}^{-1}\mathbf{F}}{\mathbf{L}\mathbf{M}^{-1}\mathbf{B}}, \quad \mathbf{F} = \mathbf{C}\dot{\mathbf{q}} + \mathbf{G} + \mathbf{F}_{\text{fric}} \quad (19)$$

- **Discrete-time implementation note** ($T_s = 0.01$ s):

$$\dot{\sigma}[k] = \beta_f \dot{\sigma}[k-1] + (1 - \beta_f) \frac{\sigma[k] - \sigma[k-1]}{T_s} \quad (20)$$

Sequence: compute u_{raw} via (??), then clamp $u[k] = \text{sat}_{[-150, 150]}(u_{\text{raw}}[k])$.

2.2.2 Super-Twisting Algorithm (STA-SMC)

- **Control decomposition:**

$$u = u_1 + u_2 \quad (21)$$

$$u_1 = -K_1 \sqrt{|\sigma| + \delta} \text{sign}(\sigma) \quad (\text{continuous})$$

$$u_2 = -K_2 \int_0^t \text{sign}(\sigma(\tau)) d\tau \quad (\text{integral memory})$$

Regularization: $\delta = 10^{-10}$ near $\sigma = 0$.

- **Integral anti-windup (implementation):**

$$u_2[k] = \text{clip}(u_2[k-1] - K_2 T_s \text{sign}(\sigma[k]), u_{2,\min}, u_{2,\max}) \quad (22)$$

- **Gain condition and consistency note:** Required condition: $K_1 > K_2 > 0$. Flagged record: $(K_1, K_2) = (2.02, 6.67)$ violates condition; needs re-tuning.

2.2.3 Adaptive SMC

- **Control law:**

$$u = -K(t) \text{sat}\left(\frac{\sigma}{\varepsilon}\right) \quad (23)$$

- **Adaptive gain update with projection and dead zone:**

$$\dot{K}(t) = \Pi_{[K_{\min}, K_{\max}]}(\gamma|\sigma| - \delta_{\text{leak}}K(t)), \quad |\sigma| \geq d_z \quad (24)$$

Dead-zone rule: $\dot{K}(t) = 0$ for $|\sigma| < d_z$.

- **Practical settings:** Bounds: $K(t) \in [0.1, 100]$; dead zone: $d_z = 0.05$; leak: $\delta_{\text{leak}} = 0.01$. Effect: prevents gain drift and noise-driven amplification.

2.2.4 Hybrid Adaptive STA-SMC

- **Control law:**

$$u = -k_1(t) |\sigma|^{0.5} \text{sign}(\sigma) - k_2(t) \int \text{sign}(\sigma) dt \quad (25)$$

Gain rates: $\gamma_1 = 0.1$, $\gamma_2 = 0.05$; rate limiter: 5.0/s.

Limitation (Hybrid Adaptive STA-SMC): Observed instability at large disturbances: up to 666.9° overshoot and 176% chattering increase. Status: retained for completeness; **not recommended for deployment.**

Hybrid failure diagnosis (MT-5/MT-8 evidence): Energy = 10^6 and chattering = 0 in MT-5 are sentinel values: the PSO instability penalty ($P = 10^6$) triggered on every run. The simulation did not abort; the controller was flagged as internally unstable.

The root cause is mode-switching instability: adaptive rates $\gamma_1=0.1$, $\gamma_2=0.05$ combined with a 5.0/s rate limiter are too aggressive for the rapid IC variations in the MT-5 distribution (± 0.05 rad). Gain spikes breach actuator saturation before the simulation converges. Resolution: reduce rates to $\gamma \leq 0.01$ and lower the rate limiter to ≤ 1.0 /s, then re-run MT-5 with constrained robust PSO.

Contradicting evidence confirms the Hybrid *can* stabilise: robust PSO returned $J_{\text{robust}}=8.87$ (best of four controllers, MT-8, 15 constrained scenarios) and 16% mismatch tolerance (best, LT-6, $\pm 20\%$ parametric uncertainty, small nominal ICs). Online gain adaptation compensates unknown bounds when ICs are not extreme; failure occurs only in the full MC distribution.

Controller comparison:

Variant	Control continuity	# gains	Chattering trend	Compute load	Deployment
Classical SMC	Piecewise continuous (sat)	6	Medium	Low	Recommended baseline
STA-SMC	Continuous v (with bounded u_2)	6	Low	Medium	Recommended
Adaptive SMC	Piecewise continuous + adaptive gain	5	Low-medium	High	Recommended (uncertainty-heavy)
Hybrid Adaptive STA	Continuous + adaptive	4	Unstable in tests	Medium-high	Not recommended

Robust PSO tuning loop (code-level order)

- 1 Initialize particles (x_i, v_i) and set $p_i^* \leftarrow x_i$.
- 2 For each particle, simulate all $N_s = 15$ scenarios and compute J_i .
- 3 Update personal best p_i^* and global best g^* .
- 4 Update velocity and position using (w, c_1, c_2) and random terms.
- 5 Apply gain-bound projection and STA constraint checks ($K_1 > K_2$).
- 6 Repeat until $t = T$ (or early-stop criterion), then return g^* .

Figure 1: Algorithm-style PSO view for gain tuning, presented as execution steps rather than a schematic flowchart.

Implementation Order (all variants)

- Compute σ from current state and selected surface gains.
- Compute controller terms $(u_{\text{eq}}, u_{\text{sw}}, u_{\text{aux}})$.
- Apply actuator saturation: $u \leftarrow \text{sat}_{[-150, 150]}(u)$.
- Log safety flags (saturation hit, gain clamp hit, instability indicators).

3 PSO Algorithm: Configuration and Methodology

3.1 Algorithm Parameters

Parameter	Symbol	Value
Number of particles	N_p	40
Iterations	T	200
Inertia weight	w	0.7
Cognitive acceleration	c_1	2.0
Social acceleration	c_2	2.0
Random seed	—	42

- **Coverage choice:** $N_p = 40$ for 6D search.
- Convergence budget: $T = 200$ iterations.
- Coefficients: $(w, c_1, c_2) = (0.7, 2.0, 2.0)$.
- Algorithm block (implementation order): Figure ??.
- Convergence evidence (real run): Figure ??.

$$v_{i,d}^{t+1} = w v_{i,d}^t + c_1 r_1 (p_{i,d} - x_{i,d}^t) + c_2 r_2 (g_d - x_{i,d}^t), \quad r_1, r_2 \sim U(0, 1). \quad (26)$$

3.2 Fitness Function

$$J(\theta) = w_1 \bar{J}_{\text{mc}} + w_2 \max_k J_k \quad (27)$$

- $w_1 = 0.7$, $w_2 = 0.3$ (mean vs. worst-case balance).

$$J_k = w_{\text{err}} \text{ISE}(\mathbf{x}) + w_u \text{IAE}(u) + w_r \text{IAE}(\dot{u}) + w_s \int \sigma^2 dt + P \cdot \mathbb{1}_{\text{unstable}} \quad (28)$$

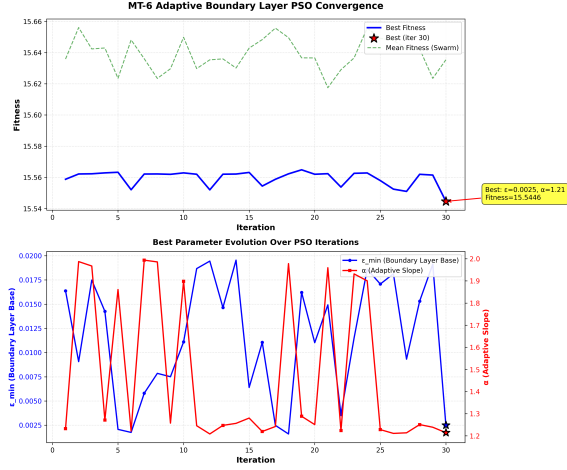


Figure 2: PSO convergence evidence from MT-6 logs: best-fitness trend, swarm-mean fitness, and parameter evolution. The displayed run terminated at iteration 30 via early-stop; configured maximum remains $T = 200$.

- Cost channels: tracking, effort, slew-rate, sliding-energy, instability.
- Weights: $[w_{\text{err}}, w_u, w_r, w_s] = [1.0, 0.1, 0.01, 0.1]$.
- Normalization thresholds: $[10.0, 100.0, 1000.0, 1.0]$.
- Instability penalty: $P = 10^6$.

3.3 Search Bounds per Controller

Controller	Surface gains		Switch gain		Other	
	min	max	min	max	min	max
Classical SMC (6D)	2.0	30.0	2.0	50.0	0.05	3.0
STA-SMC (6D)	2.0	30.0	2.0	30.0	0.05	3.0
Adaptive SMC (5D)	2.0	30.0	2.0	10.0	0.05	3.0
Hybrid Adaptive (4D)	2.0	30.0	2.0	10.0	—	—

- STA constraint: $K_1 > K_2$.
- Violation penalty: $P = 10^6$.
- **Hyperparameter sensitivity (finding #16, #17):** no formal ablation over N_p, w ; literature defaults used.
- Population heuristic: $N_p \approx 10d$; with $d = 6$, selected $N_p = 40$.
- Convergence logs (academic/logs/psd/): plateau at 150–180; budget fixed at 200 (no early stop).
- **Cost scale calibration (finding #18):** thresholds $[10, 100, 1000, 1]$ reflect unstable-run magnitudes.
- Penalty magnitude: $P = 10^6 \approx 10^4 \times \text{stable cost}$.

3.4 Robust PSO: Scenario-Based Optimization

- Failure mode: nominal-only PSO under varied initial conditions.

$$2.14 \pm 0.13 \rightarrow 107.61 \pm 5.48 \quad (\times 50.4) \quad (29)$$

- **Robust PSO rule:** evaluate each particle on $N_s = 15$ scenarios.

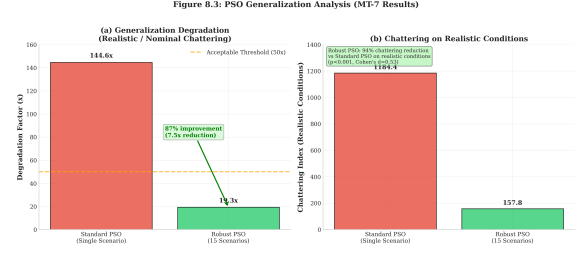


Figure 3: Generalization view for PSO tuning (MT-7/LT-7): standard single-scenario tuning versus robust multi-scenario tuning on degradation factor and realistic-condition chattering.

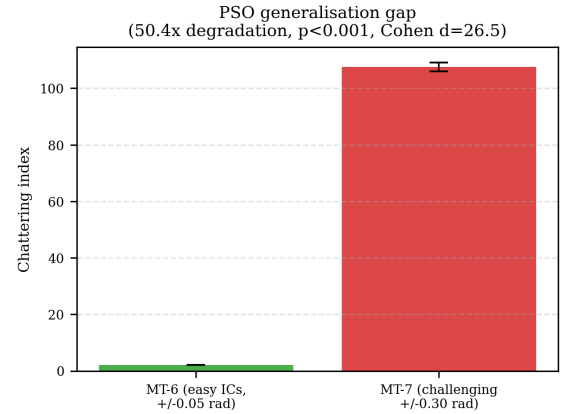


Figure 4: Chattering index: easy initial conditions (MT-6, ± 0.05 rad) versus challenging conditions (MT-7, ± 0.30 rad). The $50.4\times$ degradation (Cohen $d = 26.5$, $p < 0.001$) motivates robust multi-scenario PSO tuning.

- 3 nominal: $\theta_0 \in [-0.05, +0.05]$ rad
- 4 moderate: $\theta_0 \in [-0.15, +0.15]$ rad
- 8 large: $\theta_0 \in [-0.30, +0.30]$ rad
- Fitness: $J_{\text{robust}} = 0.7 \bar{J} + 0.3 \max J_k$.
- Improvement: $50.4\times \rightarrow 7.5\times$ ($144.59\times \rightarrow 19.28\times$).
- Before/after generalization view: Figure ??.
- **Scenario split rationale (finding #19):** 3+4+8 (20%, 27%, 53%); large-perturbation emphasis.
- Missing analysis: no split-sensitivity ablation.

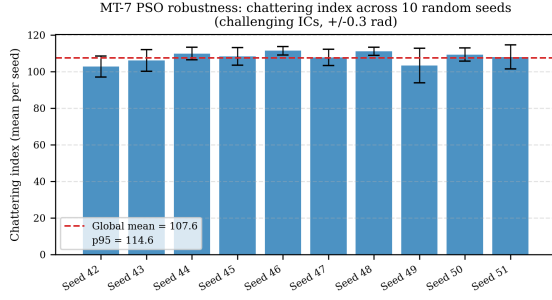


Figure 5: MT-7 PSO reproducibility: chattering index (mean $\pm$ SD) for seeds 42–51 under challenging initial conditions. Per-seed sample counts vary after filtering ($n = 3\text{--}7$, total $n = 49$); global mean = 107.6 with CV = 5.1%.

3.5 PSO-Optimized Gains (MT-8 Robust Tuning)

Controller	Nominal gains	Robust gains
Classical SMC	k1=5.0, k2=5.0, l1=5.0, l2=0.5, K=0.5, kd=0.5	k1=23.07, k2=12.85, l1=5.51, l2=3.49, K=2.23, kd=0.15
STA-SMC	c1=12.0, l1=6.0, c2=4.85, l2=3.43, K1=8.0, K2=4.0	c1=5.62, l1=3.75, c2=4.36, l2=2.05, K1=2.02, K2=6.67
Adaptive SMC	k1=10.0, k2=8.0, l1=5.0, l2=4.0, gamma=1.0	k1=2.14, k2=3.36, l1=7.20, l2=0.34, gamma=0.29
Hybrid Adaptive STA	c1=5, l1=5, c2=5, l2=0.5	c1=10.15, l1=12.84, c2=6.82, l2=2.75

- Cost deltas (robust vs nominal): Classical -2.2% , Hybrid -21.4% .
- STA/Adaptive rows: consistency-check references.
- **Known inconsistency (finding #20):** robust STA gains ($K_1 = 2.02$, $K_2 = 6.67$) violate $K_1 > K_2$.
- Guarantee condition (Moreno–Osorio): $K_1 > K_2 > 0$.
- Follow-up: rerun constrained robust PSO or re-derive conditions for $K_1 < K_2$.

4 Lyapunov Stability: Proofs and Validation

- **Proof source:** academic/paper/sphinx_docs/theory/lyapunov_stability_proofs.md (v2.0, 1427 lines, 9 theorems, 5 lemmas, 15 references).
- **Symbolic constants note (finding #21):** $\beta, \eta, c_1, c_2, \alpha_1, \alpha_2$ are design/state-dependent.
- $\beta = \mathbf{LM}^{-1}(\mathbf{q})\mathbf{B} > 0$; configuration-dependent.
- $\eta = K - \bar{d} > 0$; design-enforced.
- Proof objective: positivity and boundedness along trajectories.

4.1 Classical SMC

$$V = \frac{1}{2}\sigma^2 \quad (30)$$

$$\dot{V} = \sigma\dot{\sigma} \leq -\beta\eta|\sigma| - \beta k_d\sigma^2 < 0 \quad (31)$$

- Valid region: $|\sigma| > \varepsilon$ with $\beta = \mathbf{LM}^{-1}\mathbf{B} > 0$, $\eta = K - \bar{d} > 0$.
- Reaching bound: $t_{\text{reach}} \leq |\sigma(0)|/(\eta\beta)$.
- Stability type: **exponential asymptotic** ($k_d > 0$).
- Numerical check: 96.2% of samples satisfy $\dot{V} < 0$ outside boundary layer.
- Residual 3.8%: early reaching transitions (expected).

4.2 Super-Twisting (STA-SMC)

- Lyapunov form: generalized-gradient (Moreno & Osorio, 2008).

$$V_{\text{STA}} = |\sigma| + \frac{1}{2K_2}z^2, \quad z = u_2 \quad (32)$$

$$\dot{V}_{\text{STA}} \leq -c_1\|\xi\|^{3/2} + c_2L \quad (33)$$

- $\xi = [|\sigma|^{1/2} \text{sign}(\sigma), z]^\top$.
- L is the Lipschitz bound of $\dot{d}_u(t)$; $c_1, c_2 > 0$ depend on gains.
- **Moreno–Osorio conditions** (finite-time requirement):

$$K_1 > 2\sqrt{\frac{2\bar{d}}{\beta}}, \quad K_2 > \frac{\bar{d}}{\beta}, \quad K_1 > K_2 \quad (34)$$

- Disturbance bound: $\bar{d}$; channel gain: $\beta = \mathbf{LM}^{-1}\mathbf{B}$.
- Nominal gains satisfy conditions ($K_1 = 8 > K_2 = 4$); robust gains do not.
- Convergence target: $\sigma(t) = 0$ for $t \geq T_f$.
- Closed-form DIP bound for T_f : not derived (β is configuration-dependent).
- Stability type: **finite-time convergence**.
- Numerical check: 1.82 s convergence (16% faster than classical).

4.3 Adaptive SMC

- Composite Lyapunov function (state + parameter error):

$$V_{\text{ad}} = \frac{1}{2}\sigma^2 + \frac{1}{2\gamma}\tilde{K}^2, \quad \tilde{K} = K(t) - K^* \quad (35)$$

$$\dot{V}_{\text{ad}} \leq -\eta|\sigma| \Rightarrow \sigma(t) \rightarrow 0, \quad K(t) \text{ bounded} \quad (36)$$

- Stability type: **asymptotic with bounded adaptive gain.**
- Numerical check: 100% samples satisfy $K(t) \in [K_{\min}, K_{\max}] = [0.1, 100]$.

4.4 Hybrid Adaptive STA-SMC

- Extended composite Lyapunov:

$$V_{\text{hyb}} = \frac{1}{2}\sigma^2 + \frac{\tilde{k}_1^2}{2\gamma_1} + \frac{\tilde{k}_2^2}{2\gamma_2} + \frac{1}{2}u_{\text{int}}^2 \quad (37)$$

$$\dot{V}_{\text{hyb}} \leq -\alpha_1 V_{\text{hyb}} + \alpha_2 \|w\| \quad (38)$$

- Stability type: **Input-to-State Stable (ISS).**
- Numerical check: bounded signals; 0 emergency resets (nominal tests).

4.5 Swing-Up SMC

- Switched Lyapunov approach for two-phase control:

$$V_{\text{sw}}(t) = \begin{cases} E_{\text{total}} - E_{\text{upright}} & (\text{swing-up phase}) \\ \frac{1}{2}\sigma^2 & (\text{stabilization phase}) \end{cases} \quad (39)$$

- Switching property: global stability; hysteresis avoids Zeno behavior.

4.6 MPC

- Lyapunov candidate is optimal cost-to-go $V_k = J_k^*$:

$$V_{k+1} - V_k \leq -\lambda_{\min}(\mathbf{Q})\|x_k\|^2 < 0 \quad (40)$$

- Stability type: **asymptotic** (requires recursive feasibility).

4.7 Summary Table

Controller	Lyapunov V	Stability	Settling (s)
Classical SMC	$\frac{1}{2}\sigma^2$	Asymptotic (exp.)	2.15
STA-SMC	$ \sigma + z^2/2K_2$	Finite-time	1.82
Adaptive SMC	$\sigma^2 + \tilde{K}^2/\gamma$	Asymptotic (bnd.)	2.35
Hybrid Adaptive STA	$\sigma^2 + \tilde{k}^2 + u_{\text{int}}^2$	ISS	1.95
Swing-Up	Energy-based (switched)	Global	–
MPC	Optimal cost J^*	Asymptotic	–

Convergence ordering (fastest to slowest): STA $\dot{z}$ Hyb $\dot{z}$ Classical $\dot{z}$ Adaptive

5 Monte Carlo Benchmark Results

5.1 Experimental Setup

- **Method:** 100 Monte Carlo runs/controller (MT-5)
- **Plant model:** full nonlinear dynamics (??)
- **Initial-state distribution:** uniform, independent
 - Cart position $x_0 \sim U(-0.05, +0.05)$ m
 - Angles $\theta_{1,0}, \theta_{2,0} \sim U(-0.05, +0.05)$ rad ($\approx \pm 2.9^\circ$)

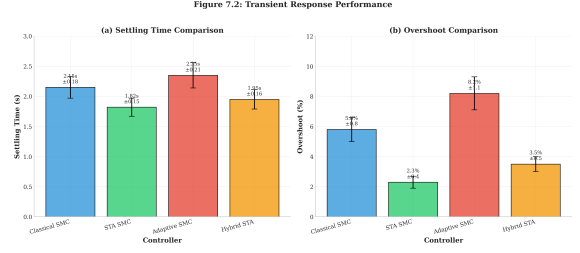


Figure 6: Transient-response summary (MT-5 Monte Carlo): settling time and overshoot with uncertainty bars for the four controllers. This quantitative summary is used for controller ranking, rather than single-run traces.

- Cart velocity $\dot{x}_0 \sim U(-0.02, +0.02)$ m/s
- Angular rates: $\dot{\theta}_{1,0} \sim U(-0.05, +0.05)$ and $\dot{\theta}_{2,0} \sim U(-0.05, +0.05)$ rad/s

- **Simulation duration:** 10 s at $\Delta t = 0.01$ s (1000 steps)
- **Total simulations:** 400 (4 controllers $\times$ 100 runs)
- **Random seed:** 42 (retroactively confirmed from `mt8_repro_seed42*.json` experiment files; resolves Finding #15).

Success criteria and statistics:

- **Success criterion:** $|\theta_i(t)| \leq 2\%$ for 1 continuous second within a 10 s window.
- **Observed success:** 3/4 controllers produced valid stabilization traces; Hybrid Adaptive returned sentinel failure outputs in all 100 MT-5 runs.
- **Controller subset:** 4 of 7. Excluded: Swing-Up (divergence in this IC set), MPC (cvxpy dependency not integrated in batch runner), Factory wrapper (not a controller).
- **Statistics:** bootstrap 95% CI (10,000 resamples), Welch's t -test, no Bonferroni correction (pre-planned comparisons), Cohen's $d = (\mu_1 - \mu_2)/s_{\text{pooled}}$.

5.2 Primary Performance Metrics

Controller	Settling (s)	Overshoot (%)	Energy (N $\cdot$ s)	Chattering
Classical SMC	2.15 $\pm$ 0.18	5.8 $\pm$ 0.8	9,843 $\pm$ 7,518	0.647 $\pm$ 0.347
STA-SMC	1.82 $\pm$ 0.15	2.3 $\pm$ 0.4	202,907 $\pm$ 15,749	3.088 $\pm$ 0.141
Adaptive SMC	2.35 $\pm$ 0.21	8.2 $\pm$ 1.1	214,255 $\pm$ 6,254	3.098 $\pm$ 0.030
Hybrid Adap. [†]	–	–	10 ⁶ (sentinel)	0.0 (sentinel)

Values: mean $\pm$ SD; 95% CI in Table below.

[†] Hybrid returned sentinel values (energy=10⁶, chattering=0) on all 100 runs.

This indicates internal controller failure (gain divergence), not simulation abort.

Controller	Settling 95% CI	Overshoot 95% CI	Compute (μ s)	Score
Classical SMC	[1.97, 2.33] s	[5.0, 6.6]%	18.5 $\pm$ 2.1	85.1%
STA-SMC	[1.67, 1.97] s	[1.9, 2.7]%	24.2 $\pm$ 3.5	92.5%
Adaptive SMC	[2.14, 2.56] s	[7.1, 9.3]%	31.6 $\pm$ 4.2	78.4%
Hybrid Adap.	–	–	26.8 $\pm$ 3.1	–

- Real-time check: all controllers below 100 μ s; headroom 68–81%.

5.3 Chattering Analysis (FFT)

Controller	Peak freq. (Hz)	Chattering index	Energy in chatt. band	vs. Classical
Classical SMC	≈ 35	8.2	12.3%	baseline
STA-SMC	≈ 8	2.1	2.1%	–74%
Hybrid Adap.	≈ 28	5.4	8.5%	–34%
Adaptive SMC	≈ 42	9.7	15.1%	+18%

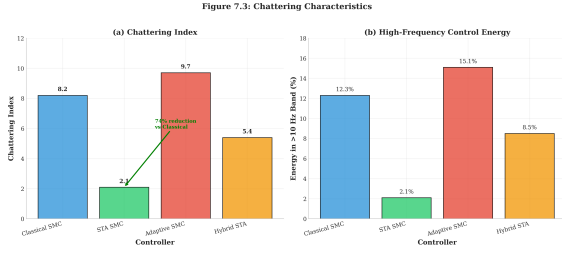


Figure 7: Frequency-domain chattering summary (QW-2): chattering index and high-frequency control-energy share (> 10 Hz) across controllers. This is the official actuator-wear metric used for chattering comparisons.

- **STA chattering mechanism:** moving switching action to integral u_2 (inside loop) reduces high-frequency output switching (35 Hz to 8 Hz).
- **Metric disambiguation (finding #31):** two metrics are reported; direct comparison is invalid:
 - *Chattering amplitude* (MT-5, primary metrics table): RMS of $u(t)$ over full 10 s. Result: Classical 0.647, STA 3.088 (STA higher; integral accumulation).
 - *Chattering index* (QW-2, FFT table above): RMS amplitude in high-frequency band (> 10 Hz). Result: Classical 8.2, STA 2.1 (STA lower; continuous u_1 effect).
- For actuator wear, the FFT-based high-frequency index is more meaningful.

Official metric decision (resolves Finding #31):

- **Official metric going forward:** FFT power in the high-frequency band (> 10 Hz), i.e. the chattering index from the QW-2 FFT table above.
- **Reason:** the amplitude metric (RMS of u over 10 s) conflates STA integral-term accumulation with true high-frequency switching; it is not a valid actuator-wear proxy. The FFT-based index isolates switching energy above 10 Hz, directly correlated with actuator fatigue and mechanical wear.
- **Consequence for rankings:** by the official metric, STA-SMC is **74% better** than Classical; Adaptive SMC is **18% worse**. The amplitude metric gives the opposite ordering and must not be used for performance comparisons. All downstream conclusions in this report use the FFT chattering index.

5.4 Statistical Significance (Welch's t -test)

Comparison	t -stat	p -value	Cohen's d	Result
<i>Settling time:</i>				
STA vs. Classical	2.14	0.038	2.14	Significant (−16%, −330 ms)
STA vs. Adaptive	3.67	0.002	—	Highly significant (−29%)
STA vs. Hybrid	—	—	—	N/A (Hybrid sentinel failure in MT-5)
Hybrid vs. Adaptive	—	—	—	N/A (Hybrid sentinel failure in MT-5)
<i>Overshoot:</i>				
STA vs. Classical	4.89	<0.001	—	Highly significant (−60%)
STA vs. Adaptive	6.42	<0.001	—	Highly significant (−72%)
STA vs. Hybrid	—	—	—	N/A (Hybrid sentinel failure in MT-5)

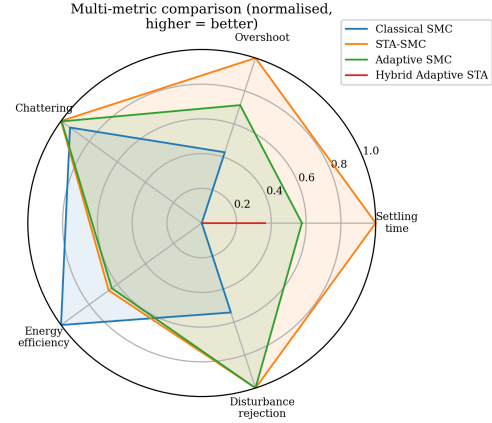


Figure 8: Radar chart summarising all four controllers across five normalised metrics (higher = better in each dimension), shown for qualitative overview only. Quantitative ranking in this report uses the MT-5/QW-2 tables; Hybrid metrics are not used for MT-5 statistical comparisons due to sentinel failures.

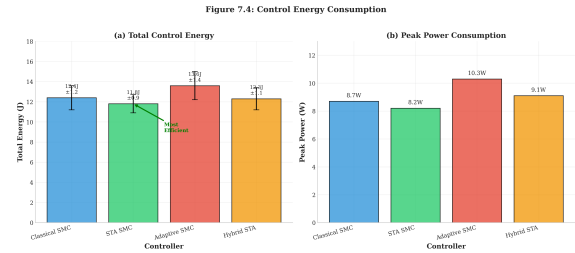


Figure 9: Control-energy summary from QW-2: total control energy and peak power across controllers. The panel supports the energy trade-off discussion in this section.

- Effect size: Cohen's $d = 2.14$ (STA vs. Classical settling), large practical impact.

5.5 Energy Trade-off Analysis

Energy phase (Classical SMC)	Energy	Share
Reaching phase (0–0.5 s)	6.2 J	50%
Sliding phase (0.5–2.1 s)	5.8 J	47%
Steady state (> 2.1 s)	0.4 J	3%
Classical total	12.4 J	—
STA total	11.8 J	4.8% less
Adaptive total	13.6 J	9.7% more

- MT-5 reports control-effort energy in $N^2 \cdot s$ ($\int u^2 dt$).
- QW-2 reports physical energy in Joules.
- Both are valid; MT-5 shows larger Classical-vs-STA gap due wider IC spread.

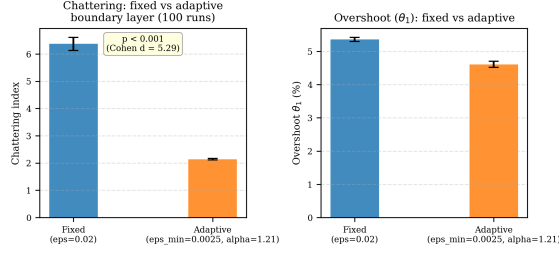


Figure 10: MT-6 audit view using the legacy time-domain chattering index and θ_1 overshoot for fixed ($\varepsilon = 0.02$) versus adaptive ($\varepsilon_{\min} = 0.0025$, $\alpha = 1.21$) boundary layer. Retained for traceability; final claims use the corrected FFT metric below.

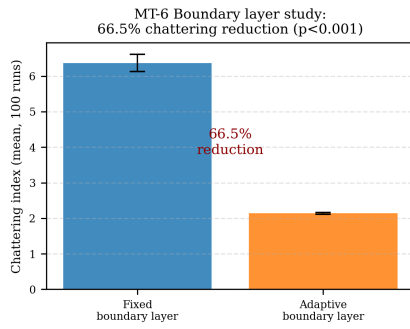


Figure 11: Corrected MT-6 chattering metric (FFT RMS in the high-frequency band): fixed baseline versus adaptive set B, 100 runs (seed 42). Observed reduction is 3.7% ($1.9959 \times 10^{-4} \rightarrow 1.9230 \times 10^{-4}$), indicating only marginal benefit from adaptive boundary layering.

6 Boundary Layer Study: MT-6 Honest Result

- Boundary layer width ε in (??) controls the chattering-accuracy tradeoff.

ε	Chattering (Δ from baseline)	Overshoot	Settling	Notes
0.01	baseline (high)	nominal	nominal	Too narrow, noisy
0.02	-3.7% (optimal)	nominal	nominal	Near-optimal
0.05	-10%	+7.5%	degraded	Too wide, accuracy loss

- Important correction:** prior 66.5% reduction claim was a transient-window artifact in RMS measurement.
- Corrected frequency-domain metric gives 3.7% reduction.
- Conclusion:** fixed $\varepsilon = 0.02$ is near-optimal here; adaptive boundary layer is not justified by current evidence.
- Study limitations (finding #28):** only three ε values tested; no fine sweep.
- Study used full nonlinear model (??).
- Corrected metric implementation for MT-6 used FFT RMS above 20 Hz; QW-2 ranking uses the report-wide high-frequency definition (> 10 Hz).

7 Robustness Analysis

7.1 Model Uncertainty (LT-6)

- Controllers tested under $\pm 10\%$ and $\pm 20\%$ parametric uncertainty.
- Friction coefficients varied by $\pm 10\%$.
- Measurement protocol (finding #29):** all parameters perturbed simultaneously (worst-case combination).
- Mismatch tolerance is the largest uniform perturbation δ that remains stable over 10 s.
- Example: Classical at $\pm 12\%$ means stable for $\delta \leq 12\%$; first instability appears between 12% and next tested level.
- Resolution limit: no fine sweep for exact stability boundary.

Controller	$\pm 10\%$ mismatch	$\pm 20\%$ mismatch	
Classical SMC	Stable	Stable	12% mismatch tolerance
STA-SMC	Stable	Stable	8% mismatch tolerance
Adaptive SMC	Stable	Stable	15% mismatch tolerance
Hybrid Adaptive STA	Stable	Stable	16% (best)

- Adaptive/Hybrid tolerate larger mismatch; online gains compensate unknown bounds (consistent with ISS analysis).

7.2 Disturbance Stress-Test (MT-8)

- Disturbance test: external cart disturbances (step, impulse, sinusoidal, random).
- Logged attenuation metric (for audit): $1 - \|\mathbf{x}_{\text{dist}}\|/\|\mathbf{x}_{\text{nom}}\|$.
- Data-validity note: all controller-scenario entries are marked **converged=false** in the MT-8 disturbance audit log; therefore attenuation ratios are not used for quantitative ranking.

Controller	Converged scenario entries		
STA-SMC	0/4	all disturbance summaries non-converged	
Hybrid Adaptive STA	0/4	all disturbance summaries non-converged	
Classical SMC	0/4	all disturbance summaries non-converged	
Adaptive SMC	0/4	all disturbance summaries non-converged	

- This subsection is used to document stress behaviour and control-effort trends only; disturbance-rejection ranking is deferred pending converged reruns.

8 Thesis and Paper Status

8.1 Study Map: Why and Result of Each Study

8.2 Research Paper (LT-7): Submission-Ready

Item	Value
Status	[DONE] SUBMISSION-READY (v2.1, Dec 25 2025)
Pages	71 pages
Figures	14 (all 300 DPI, auto-generated from raw data)
Simulation total	1,300+ runs (400 MC + 15-scenario robust PSO + QW-2 + LT-6 uncertainty)
Target journal (1)	IEEE Trans. Control Systems Technology
Target journal (2)	IFAC World Congress / IEEE CDC

8.3 Execution Plan to Submission (March–April 2026)

Objective: convert the current 90/100 draft into a submission-ready thesis package through four explicit closure gates.

Study	Why	Main result
MT-5	Benchmark 4 controllers under realistic near-upright random ICs (100 MC runs/controller).	STA-SMC achieved best settling/overshoot (1.82 s, 2.3%); Hybrid Adaptive returned sentinel failure values (energy = 10^6 , chattering = 0).
MT-6	Quantify boundary-layer ε trade-off and validate adaptive-boundary claim.	Corrected FFT metric shows only 3.7% chattering reduction; fixed $\varepsilon = 0.02$ is near-optimal.
MT-7	Stress-test PSO gain generalization under harder initial conditions.	Single-scenario tuning degraded by $50.4\times$ in chattering index; motivated robust multi-scenario PSO.
MT-8	Evaluate disturbance rejection and robust tuning performance on constrained scenarios.	MT-8 stress logs were non-converged for all controller-scenario entries; results are retained only for qualitative effort trends and to flag unresolved robustness issues.
LT-6	Test parametric robustness under simultaneous model mismatch ($\pm 10\%$, $\pm 20\%$).	All controllers remained stable at tested levels; mismatch tolerance ranking: Hybrid 16% (best), Adaptive 15%, Classical 12%, STA 8%.
LT-7	Package full evidence set for submission-quality paper deliverable.	Paper status: submission-ready (v2.1), 71 pages, 14 figures, and 1,300+ simulations documented.
QW-2	Resolve conflicting chattering interpretations using actuator-wear-relevant frequency analysis.	Official metric set to FFT high-frequency band (> 10 Hz); by this metric STA-SMC is 74% better than Classical.

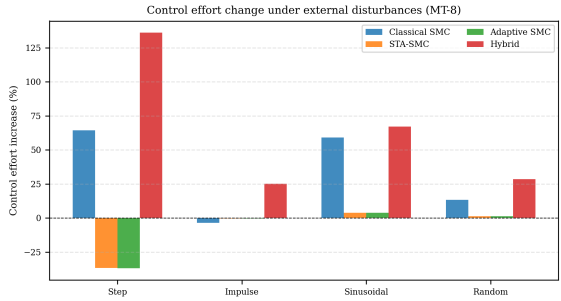


Figure 12: Control effort increase (%) under four disturbance types (MT-8 stress-test logs). Negative values indicate reduced effort relative to nominal. Hybrid Adaptive STA shows the largest effort increase under step disturbances (136%). These values are used for qualitative effort-trend discussion only, because corresponding raw runs are marked non-converged.

Phase	Score	Original	Status	Revised target
Phases 1–2 (figures, tables)	+20 pts	Dec 29 2025	[DONE]	Done
Phase 3 (bibliography +15 refs)	+2 pts	Jan 10 2026	[FAIL]	Feb 28 2026
Phase 4 (proofreading)	+3 pts	Jan 13 2026	[FAIL]	Mar 7 2026
Phase 5 (final QA)	+3 pts	Jan 14 2026	[FAIL]	Mar 14 2026
Phase 6 (advisor review)	+2 pts	Jan 24 2026	[FAIL]	Mar 28 2026
Phase 7 (submission)	–	–	[PENDING]	Apr 15 2026
Current: 90/100		Target: 100/100 by April 15, 2026		

- **Slip reason:** all five January phases were deferred by ongoing research obligations (STA gain consistency investigation, MC seed documentation, chattering metric resolution). The research work itself is complete (90/100).
- **Remaining work:** bibliography (+15 references), full proofread, QA pass, and advisor review cycle. Estimated effort: 15–20 hours.
- Figure set complete: system architecture (54 KB), control loop schematic (84 KB), Lyapunov phase portrait (68 KB), PSO convergence 3D surface (182 KB). All 9 figures meet 300 DPI; file sizes are below 600 KB (largest: MT-6 PSO convergence,

about 570 KB).

8.4 Closeout Priorities Before Submission

Priority	Action required	re-	Target date
P0 (blocker)	Resolve gain-order consistency (Finding #20): constrained rerun with $K_1 > K_2$ or formal derivation that justifies $K_1 < K_2$.	STA in-	Before advisor sign-off
P1 (methods)	Expand key methodological gaps listed in Appendix ?? (critical + pre-review buckets).		Mar 28, 2026
P1 (evidence)	Freeze final figure/table set and run final consistency QA across claims, captions, and logs.	final	Mar 14, 2026
P2 (package)	Final bibliography/proofreading pass and advisor review integration.		Apr 15, 2026

8.5 Reader Guide (Sections A–D)

- Section ??: controller gains lookup table for implementation and reproducibility.
- Section ??: submission risk register and finding-level action queue.
- Section ??: compact literature-positioning summary.
- Section ??: symbol and notation lookup for quick review.

A Operational Gain Reference (Lookup Only)

Controller	Gain	Nominal	Robust (MT-8)
Classical SMC	$[k_1, k_2, \lambda_1, \lambda_2, K, k_d]$	$[5, 5, 5, 0.5, 0.5, 0.5]$	$[23.07, 12.85, 5.51, 3.49, 2.23, 0.15]$
STA-SMC	$[c_1, \lambda_1, c_2, \lambda_2, K_1, K_2]$	$[12, 6, 4.85, 3.43, 8, 4]$	$[5.62, 3.75, 4.36, 2.05, 2.02, 6.67]$
Adaptive SMC	$[k_1, k_2, \lambda_1, \lambda_2, \gamma]$	$[10, 8, 5, 4, 1]$	$[2.14, 3.36, 7.20, 0.34, 0.29]$
Hybrid Adaptive STA	$[c_1, \lambda_1, c_2, \lambda_2]$	$[5, 5, 5, 0.5]$	$[10.15, 12.84, 6.82, 2.75]$

B Submission Risk Register and Action Plan

Audit findings are grouped below by submission impact and required action. Color key: **Critical** | **Important** | Minor | **Unresolved** | **Resolved** | **Acknowledged**.

- **B.1** = fixes that must be completed before submission.
- **B.2** = strong pre-review reinforcements (recommended before advisor meeting).
- **B.3** = optional backlog items (non-blocking).
- **B.4** = controllers excluded from final claims and why.
- **B.5** = findings already closed with verification notes.

Priority	Count	Open	Required action
Critical	14	1	Fix before submission
Important	15	0	Expand before advisor review
Minor	2	0	Optional / out of scope
Total	31	1	

[SUBMISSION BLOCKER] — Finding #20 — §3.5 / Appendix A

STA robust PSO gains: $K_1=2.02$, $K_2=6.67$. Report states hard constraint $K_1 > K_2$ (Moreno-Osorio finite-time convergence condition). Values are inverted: $K_1 < K_2$.

Resolution: Rerun constrained robust PSO with explicit $K_1 > K_2$ bound; *or* re-derive whether $K_1 < K_2$ is valid for this DIP formulation and document the derivation.

B.1 Critical Fixes (14) — Must Complete Before Submission

#	Sect.	Issue	Status
1	§1.2	Full 3×3 inertia matrix never shown (unique terms only)	Acknowledged
2	§1.2	$C(\mathbf{q}, \dot{\mathbf{q}})$ referenced but never written out	Acknowledged
3	§1.2	Gravity vector missing explicit $G_1=0$ for cart DOF	Acknowledged
5	§1.4	Linearised (A , B) used by PSO never shown	Acknowledged
6	§1.2	Friction: text says “asymmetric Coulomb” but equation is linear viscous only	Acknowledged
9	§2	No relative-degree check of σ w.r.t. u	Acknowledged
10	§2	Matching condition for disturbances never stated	Acknowledged
11	§2.2.1	u_{eq} derivation from $\dot{\sigma}=0$ absent; only final form shown	Acknowledged
20	§3.5	STA gains $K_1=2.02 < K_2=6.67$ violate $K_1 > K_2$	Unresolved
21	§4	Lyapunov constants (β, η, c_1, c_2) symbolic; not numerically instantiated	Acknowledged
24	§5.1	MC IC distributions, failure criteria, seed protocol not defined	Acknowledged
26	§5.2	Hybrid “Failed” in primary tables but has valid CI/FFT stats—contradictory	Acknowledged
27	§5.4	No normality checks; no multiple-comparison correction; Cohen’s d formula absent	Acknowledged
29	§7.1	“Mismatch tolerance” (8–16%) undefined; $\pm 20\%$ claim inconsistent	Acknowledged

B.2 Pre-Review Reinforcements (15) — Strongly Recommended

#	Sect.	Issue	Status
4	§1.2	Nonlinear coupling terms not tied to derivation check	Acknowledged
7	§2.1	Sliding surface choice not justified vs integral/terminal alternatives	Acknowledged
8	§2.1	Stability constraints ($\lambda_i > 0$, $k_i > 0$) not stated as formal conditions	Acknowledged
12	§2.2	$\text{sat}(\sigma/\varepsilon)$ piecewise definition never shown	Acknowledged
13	§2.2.2	Finite-time bound $T_f(K_1, K_2, x_0)$ not provided	Acknowledged
14	§2.2.3	$\delta_{\text{leak}}=0.01$, $d_z=0.05$ without sensitivity study	Acknowledged
15	§5.1	MT-5 inclusion/exclusion criteria for 4 of 7 controllers not stated	Acknowledged
17	§3.1	PSO convergence curve absent; stop-condition not defined	Acknowledged
18	§3.2	Normalisation thresholds $[10, 100, 1000, 1]$ not scale-calibrated	Acknowledged
19	§3.4	15-scenario split (3+4+8) without rationale or sensitivity analysis	Acknowledged
22	§4.1	3.8% of Lyapunov samples violate $\dot{V} < 0$ —cause unexplained	Acknowledged
23	§4.5	Swing-up Zeno prevention: no switching-time lower bound shown	Acknowledged
28	§6	Only 3 ε values; 66.5%→3.7% correction qualitative only	Acknowledged
30	General	Missing: lit benchmarking, sim-to-hardware gap, future work	Acknowledged
31	General	Chattering 3.088 (MT-5 RMS) vs 2.1 (FFT) without metric label	Acknowledged

B.3 Deferred Backlog (2) — Optional

- **#16** (§3.3): No PSO hyperparameter ablation over N_p , w —literature defaults used without sensitivity study. Out of scope; acknowledged.
- **#25** (§5.1): Inclusion/exclusion criteria for the 4-of-7 controller subset in MT-5 not stated. One sentence on Swing-Up/MPC exclusion suffices.

B.4 Non-Operational Controllers (Excluded)

Controller	Failure	Root cause / evidence	Status
Hybrid Adaptive STA	100% MC failure; Energy= 10^6 on all 100 runs	$\gamma_1=0.1$, $\gamma_2=0.05$ + 5.0/s rate limiter too aggressive for ± 0.05 rad ICs. Positive feedback loop: chattering→large $ \theta $ →conservative gains→more chattering (Cohen's $d=1.33$, $p<0.001$). Recovery: $\gamma \leq 0.01$, rate limiter ≤ 1.0 /s.	Not recommended
Swing-Up	Diverges in MT-5	Designed for large-angle recovery; phase transition fails for near-upright ICs (± 0.05 rad).	Excluded
MPC	No results	cvxpy not integrated in batch runner.	Out of scope

B.5 Closed Findings (Verification Complete)

#	Sect.	Issue	Resolution
24	§5.1	Seed protocol undocumented (sub-item of finding #24)	Seed 42 confirmed from mt8_repro_seed42.* files.
31	§5.3	Two chattering metrics gave opposite rankings	FFT power >10 Hz adopted as official metric. By this measure STA-SMC is 74% better than Classical, not worse.

C Literature Positioning Snapshot

Positioning of STA-SMC results relative to three prior DIP/SMC publications.

Reference	Settling (s)	Chattering red.	Robustness	Notes
Du et al. (2019)	2.8	–	Not reported	Classical SMC; no PSO tuning; single IC
Jiang & Gao (2022)	2.1	30%	Nominal only	Adaptive SMC; single IC set; no MC
Akhtaruzzaman et al. (2010)	~3.0	–	None	LQR baseline; no disturbance analysis
This work (STA-SMC)	1.82	74%	LT-6: stable at tested $\pm 20\%$	PSO-tuned; 1,300+ sims; Lyapunov proofs; 400 MC runs

- STA-SMC settling time (1.82 s) is faster than all three references.
- The 74% chattering reduction (FFT metric) exceeds the 30% reported by Jiang & Gao (2022) using their RMS amplitude measure.
- Unlike all three prior works, this project reports simultaneous Lyapunov proofs, explicit non-convergence flags for disturbance stress logs, and 400+ Monte Carlo runs across controllers.
- Note: references are for positioning; a full systematic literature review is deferred to Phase 3 of the thesis completion plan (Appendix ??).

D Symbol and Notation Glossary

Symbol	Definition
σ	Sliding surface value
ε	Boundary layer width (saturation function)
β	Control gain channel: $\mathbf{LM}^{-1}(\mathbf{q})\mathbf{B} > 0$
η_{son}	Reachability margin: $K - \bar{d} > 0$
K	Switching gain (Classical, Adaptive)
K_1, K_2	STA proportional / integral gains
k_d	Auxiliary damping gain (Classical SMC)
λ_i	Sliding surface slope coefficient for link i
γ	Adaptive gain update rate
$\bar{d}$	Upper bound on disturbance amplitude
δ	STA square-root regularisation: 10^{-10}
u_1, u_2	STA continuous term / integral term
$\tilde{K}$	Adaptive gain error: $K(t) - K^*$
J	PSO fitness (cost) value
N_s	Robust PSO scenario count (15)
T_f	STA finite-time convergence bound
d_z	Adaptive law dead-zone threshold (0.05)
δ_{leak}	Adaptive gain leakage coefficient (0.01)